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CALCUL NUMERIC
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Factorizarea LU

Exemplu:

$$Sa\ se\ scrie\ matricea\ A = \begin{pmatrix} 2 & 3 & 1 \\ 4 & 7 & 5 \\ 2 & 5 & 10 \end{pmatrix}\ sub\ forma\ A = L * U, unde$$

$$L\ respectiv\ U\ sunt\ matrice\ inferior\ triunghiulara\ respectiv\ superior\ triunghiulara.$$

Rezolvare:

$$Dupa\ ce\ verificam\ daca\ poate\ fi\ aplicata\ factorizarea\ (vezi\ curs):$$

$$A = \begin{pmatrix} 2 & 3 & 1 \\ 4 & 7 & 5 \\ 2 & 5 & 10 \end{pmatrix} = \begin{pmatrix} a_{00} & a_{01} & a_{02} \\ a_{10} & a_{11} & a_{12} \\ a_{20} & a_{21} & a_{22} \end{pmatrix} = \begin{pmatrix} l_{00} & 0 & 0 \\ l_{10} & l_{11} & 0 \\ l_{20} & l_{21} & l_{22} \end{pmatrix} \begin{pmatrix} u_{00} & u_{01} & u_{02} \\ 0 & u_{11} & u_{12} \\ 0 & 0 & u_{22} \end{pmatrix}$$

$$Rescriem\ matricele\ pe\ blocuri:$$

$$\begin{pmatrix} a_{00} & A_{01} \\ A_{10} & A_{11} \end{pmatrix} = \begin{pmatrix} l_{00} & 0 \\ L_{10} & L_{11} \end{pmatrix} \begin{pmatrix} u_{00} & U_{01} \\ 0 & U_{11} \end{pmatrix}$$

$$Obtinem\ sistemul:$$

$$\begin{cases} a_{00} = l_{00} * u_{00} + 0 * 0 \\ A_{01} = l_{00} * U_{01} + 0 * U_{11} \\ A_{10} = L_{10} * u_{00} + L_{11} * 0 \\ A_{11} = L_{10} * U_{01} + L_{11} * U_{11} \end{cases} \Rightarrow \begin{cases} a_{00} = l_{00} u_{00} \\ A_{01} = l_{00} U_{01} \\ A_{10} = L_{10} u_{00} \\ A_{11} = L_{10} U_{01} + L_{11} U_{11} \end{cases}$$

$$Obtinem\ o\ infinitate\ de\ produse\ LU = A, deci\ alegem\ l_{00} = 1$$

$$Sistemul\ devine: \begin{cases} a_{00} = u_{00} \quad (1) \\ A_{01} = U_{01} \\ A_{10} = u_{00} L_{10} \\ A_{11} = L_{10} U_{01} + L_{11} U_{11} \end{cases}$$

$Stim\ din\ ipoteza\ faptul\ ca\ a_{00} = 2 \Longrightarrow Din\ ecuatia\ (1):\ a_{00} = u_{00} = 2$

$$Sistemul\ devine: \begin{cases} A_{01} = U_{01} & (2) \\ A_{10} = 2L_{10} & (3) \\ A_{11} = L_{10}U_{01} + L_{11}U_{11} & (4) \end{cases}$$

$Din\ ecuatia\ (2): A_{01} = U_{01} \Leftrightarrow (a_{01} \quad a_{02}) = (u_{01} \quad u_{02}) \Leftrightarrow \begin{cases} a_{01} = u_{01} \\ a_{02} = u_{02} \end{cases} \Leftrightarrow$

$\Leftrightarrow \begin{cases} u_{01} = 3 \\ u_{02} = 1 \end{cases}$

$Din\ ecuatia\ (3): A_{10} = 2L_{10} \Leftrightarrow \begin{pmatrix} a_{10} \\ a_{20} \end{pmatrix} = 2\begin{pmatrix} l_{10} \\ l_{20} \end{pmatrix} \Leftrightarrow \begin{cases} 4 = 2l_{10} \\ 2 = 2l_{20} \end{cases} \Leftrightarrow$

$\Leftrightarrow \begin{cases} l_{10} = 2 \\ l_{20} = 1 \end{cases}.$

$In\ continuare\ vrem\ sa\ aflam\ produsul\ L_{11}U_{11}. Din\ ecuatia\ (4)\ stim\ A_{11}\ si\ putem$

$afla\ L_{10}U_{01}\ (gasite\ anterior), iar\ L_{11}U_{11} = A_{11} - L_{10}U_{01}.$

$L_{10}U_{01} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}(3 \quad 1) = \begin{pmatrix} 6 & 2 \\ 3 & 1 \end{pmatrix}.$

$$L_{11}U_{11} = A_{11} - L_{10}U_{01} = \begin{pmatrix} 7 & 5 \\ 5 & 10 \end{pmatrix} - \begin{pmatrix} 6 & 2 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 2 & 9 \end{pmatrix}$$

Cine e $L_{11}U_{11}$?

$$L_{11}U_{11} = \begin{pmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{pmatrix}\begin{pmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 2 & 9 \end{pmatrix}$$

$$Obtinem: \begin{cases} l_{11}u_{11} = 1 \\ l_{11}u_{12} = 3 \\ u_{11}l_{21} = 2 \\ u_{12}l_{21} + l_{22}u_{22} = 9 \end{cases}$$

$Din\ aceleasi\ considerente\ alegem\ l_{11} = 1$

$$Sistemul\ devine: \begin{cases} u_{11} = 1 \\ u_{12} = 3 \\ u_{11}l_{21} = 2 & (5) \\ u_{12}l_{21} + l_{22}u_{22} = 9 & (6) \end{cases}$$

$Din\ ecuatia\ (5)\ u_{11}l_{21} = 2; dar\ u_{11} = 1 \Longrightarrow l_{21} = 2$

$Ecuatia\ (6)\ devine: 2 * 3 + l_{22}u_{22} = 9 \Longrightarrow l_{22}u_{22} = 3$

$Alegem\ l_{22} = 1\ deci\ u_{22} = 3$

$$Am\ obtinut:$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 2 & 1 \end{pmatrix} \qquad U = \begin{pmatrix} 2 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 3 \end{pmatrix}$$

$$Verificam: \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 2 & 1 \end{pmatrix} \begin{pmatrix} 2 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 3 \end{pmatrix} =$$

$$\begin{pmatrix} 1*2+0*0+0*0 & 1*3+0*1+0*0 & 1*1+0*3+0*3 \\ 2*2+1*0+0*0 & 2*3+1*1+0*0 & 2*1+1*3+0*3 \\ 1*2+2*0+1*0 & 1*3+2*1+1*0 & 1*1+2*3+1*3 \end{pmatrix} =$$

$$= \begin{pmatrix} 2 & 3 & 1 \\ 4 & 7 & 5 \\ 2 & 5 & 10 \end{pmatrix} = A$$

Exerciții propuse:

$$Analog\ pentru \begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 3 & 7 & 8 \end{pmatrix}, respectiv \begin{pmatrix} 3 & 1 & 2 \\ 6 & 3 & 3 \\ 3 & 2 & 7 \end{pmatrix}.$$

Factorizarea Cholesky

Exemplu:

$$Sa\ se\ scrie\ matricea\ C = \begin{pmatrix} 4 & 12 & -16 \\ 12 & 37 & -43 \\ -16 & -43 & 98 \end{pmatrix} sub\ forma\ C = LL^t; unde\ L$$

$$este\ o\ matrice\ inferior\ triunghiulara$$

Rezolvare:

$$Dupa\ ce\ verificam\ daca\ poate\ fi\ aplicata\ factorizarea\ (vezi\ curs):$$

$$\begin{pmatrix} c_{00} & c_{01} & c_{02} \\ c_{10} & c_{11} & c_{12} \\ c_{20} & c_{21} & c_{22} \end{pmatrix} = \begin{pmatrix} l_{00} & 0 & 0 \\ l_{10} & l_{11} & 0 \\ l_{20} & l_{21} & l_{22} \end{pmatrix} \begin{pmatrix} l_{00} & l_{10} & l_{20} \\ 0 & l_{11} & l_{21} \\ 0 & 0 & l_{22} \end{pmatrix}$$

$L^t\ este\ transpusa\ unei\ matrice\ inferior\ triunghiulara, deci\ e\ superior$

$triunghiulara\ (ne\ comportam\ ca\ la\ factorizarea\ LU)$

$Rescriem\ matricele\ pe\ blocuri:$

$$\begin{pmatrix} c_{00} & C_{01} \\ C_{10} & C_{11} \end{pmatrix} = \begin{pmatrix} l_{00} & 0 \\ L_{10} & L_{11} \end{pmatrix} \begin{pmatrix} l_{00} & L_{01} \\ 0 & {L_{11}}^t \end{pmatrix}$$

$$Obtinem\ sistemul: \begin{cases} c_{00} = {l_{00}}^2 & (6) \\ C_{01} = l_{00}L_{01} & (7) \\ C_{10} = l_{00}L_{10} & (8) \\ C_{11} = L_{01}L_{10} + L_{11}{L_{11}}^t & (9) \end{cases}$$

$Din\ ecuatia\ (6) \Longrightarrow l_{00} = 2$

$$Din\ ecuatia\ (7) \Longrightarrow (c_{01} \quad c_{02}) = (l_{10} \quad l_{20}) \Leftrightarrow \begin{cases} c_{01} = l_{10} \\ c_{02} = l_{20} \end{cases} \Longrightarrow \begin{cases} l_{10} = 12 \\ l_{20} = -16 \end{cases}$$

$Ecuatia\ (8)\ este\ echivalenta\ cu\ ecuatia\ (7), L_{01} = {L_{10}}^t$

$Din\ ecuatia\ (9)\ vrem\ sa\ aflam\ L_{11}{L_{11}}^t. Dar\ L_{11}{L_{11}}^t = C_{11} - L_{01}{L_{01}}^t.$

$In\ continuare\ de\ rezolvat\ asemanator\ factorizarii\ LU$

Factorizarea UL

Exerciții propuse:

$De\ incercat\ scrierea\ matricelor\ anterioare\ A\ sub\ forma\ A = UL$

$cu\ U, L\ matrice\ superior\ respectiv\ inferior\ triunghiulare.$